

Z. TRITICI IPO323 · DOSE GATE · 3 CULTURES × 2 TEMPERATURES

Dose-Finding: 4 plates

Three independent cultures, two temperatures, no drug. Same two plate layouts at 15 °C and 27 °C.

Plate = inhibitor. Row = culture. Column = dose (low → high, left → right). Every row is one culture's ladder, nothing mixed.

15 °C: nPG_15 + SHAM_15

27 °C: nPG_27 + SHAM_27

no prothioconazole

4 plates · 82 vials

OD₆₀₀ 0.0005

GATE before anything goes in a vial

1. **Each culture OD ≥ 0.10**, healthy under the scope. Log OD and age per culture.
2. **Reader calibrated**; a vial of plain air-shaken YMS reads ~7–8 mg/L.
3. nPG master fresh today; SHAM master from foil stock. Both clear.

nPG PLATE one at 15 °C (nPG_15), one at 27 °C (nPG_27)

	1	2	3	4	5	6
	ctrl	0.0125	0.025	0.05	0.075	0.1
	EtOH	6.25 mM	12.5 mM	25 mM	37.5 mM	50 mM
A	A1 CTRL 0	A2 nPG 0.0125	A3 nPG 0.025	A4 nPG 0.05	A5 nPG 0.075	A6 nPG 0.1
	10 µL EtOH	10 µL 6.25 mM	10 µL 12.5 mM	10 µL 25 mM	10 µL 37.5 mM	10 µL 50 mM
B	B1 CTRL 0	B2 nPG 0.0125	B3 nPG 0.025	B4 nPG 0.05	B5 nPG 0.075	B6 nPG 0.1
	10 µL EtOH	10 µL 6.25 mM	10 µL 12.5 mM	10 µL 25 mM	10 µL 37.5 mM	10 µL 50 mM
C	C1 CTRL 0	C2 nPG 0.0125	C3 nPG 0.025	C4 nPG 0.05	C5 nPG 0.075	C6 nPG 0.1
	10 µL EtOH	10 µL 6.25 mM	10 µL 12.5 mM	10 µL 25 mM	10 µL 37.5 mM	10 µL 50 mM
D	D1 BLANK nPG 0.1	D2	D3	D4	D5	D6
	10 µL 50 mM					

19 vials. Each well: final dose in mM (big) and the stock to pipette 10 µL of (small, under the line).
Column headers repeat the stock. Column 1 is that culture's control.

SHAM PLATE

one at 15 °C (SHAM_15), one at 27 °C (SHAM_27)

	1 0.0125 6.25 mM	2 0.025 12.5 mM	3 0.05 25 mM	4 0.1 50 mM	5 0.2 100 mM	6 0.35 175 mM
A culture 1	A1 SHAM 0.0125 10 µL 6.25 mM	A2 SHAM 0.025 10 µL 12.5 mM	A3 SHAM 0.05 10 µL 25 mM	A4 SHAM 0.1 10 µL 50 mM	A5 SHAM 0.2 10 µL 100 mM	A6 SHAM 0.35 10 µL 175 mM
B culture 2	B1 SHAM 0.0125 10 µL 6.25 mM	B2 SHAM 0.025 10 µL 12.5 mM	B3 SHAM 0.05 10 µL 25 mM	B4 SHAM 0.1 10 µL 50 mM	B5 SHAM 0.2 10 µL 100 mM	B6 SHAM 0.35 10 µL 175 mM
C culture 3	C1 SHAM 0.0125 10 µL 6.25 mM	C2 SHAM 0.025 10 µL 12.5 mM	C3 SHAM 0.05 10 µL 25 mM	C4 SHAM 0.1 10 µL 50 mM	C5 SHAM 0.2 10 µL 100 mM	C6 SHAM 0.35 10 µL 175 mM
D controls + blank	D1 CTRL c1 0 10 µL EtOH	D2 CTRL c2 0 10 µL EtOH	D3 CTRL c3 0 10 µL EtOH	D4 BLANK SHAM 0.35 10 µL 175 mM	D5	D6

■ control (cells + EtOH)
 ■ nPG (mM)
 ■ SHAM (mM)
 ■ cell-free blank

22 vials. Row D1-D3 are each culture's control on this plate (so the SHAM plate has its own controls, not borrowed from the other reader). D4 = blank at the top SHAM dose.

FILL

cell wells: 5.03 mL cells (OD 0.0005) + 20 µL P0 (DMSO) + 10 µL inhibitor stock (plain EtOH for CTRL)

blanks: 5.03 mL sterile YMS + 20 µL P0 + 10 µL inhibitor

CULTURE	DILUTE TO	FILLS
culture 1	~140 mL @ 0.0005	row A on all 4 plates + D1 on SHAM plates
culture 2	~140 mL @ 0.0005	row B on all 4 plates + D2 on SHAM plates
culture 3	~140 mL @ 0.0005	row C on all 4 plates + D3 on SHAM plates

mL culture = $0.0005 \times 140 \div \text{that culture's OD}$, top up with YMS. Three labelled bottles, never mixed. Each culture fills 26 wells (13 per temperature). Swirl between draws.

STOCKS 10 μL /well · ~1 mL each serves all four plates

FINAL IN VIAL	NPG STOCK (FROM 50 MM MASTER)	SHAM STOCK (FROM 250 MM MASTER)
0.0125 mM	6.25 mM · 125 μL + 875 EtOH	6.25 mM · 25 μL + 975 EtOH
0.025 mM	12.5 mM · 250 μL + 750 EtOH	12.5 mM · 50 μL + 950 EtOH
0.05 mM	25 mM · 500 μL + 500 EtOH	25 mM · 100 μL + 900 EtOH
0.075 mM	37.5 mM · 750 μL + 250 EtOH	—
0.1 mM	50 mM · master neat	50 mM · 200 μL + 800 EtOH
0.2 mM	—	100 mM · 400 μL + 600 EtOH
0.35 mM	—	175 mM · 700 μL + 300 EtOH

Not serial: each stock straight from its master. Foil all of them. Tip: line the stock tubes up in column order (1→6) next to the plate and go across each row.

RUN

1. Culture by culture: gate → dilute → fill its row on all four plates (+ its D-well on the SHAM plates) → next culture. All four plates into readers the same afternoon, dark.
2. Exports: **nPG_15, nPG_27, SHAM_15, SHAM_27**. Row letter = culture on every plate.
3. 27 °C turns over in ~4 days, 15 °C ~5–6. Harvest by the curves.
4. Same day: start the induced flask (27 °C + 2 mg/L drug, ~50 mL, ~OD 0.02) for the azide plate.

Read-out. Dose: highest well within ~15–20% of its row's control, at 27 °C; carry two doses forward. Split: inhibitor costing growth at 27 °C but not 15 °C, in all three rows = AOX engaged only in the heat.